AYOLEYI GBENGA-AYODEJI MARVELOUS

Aspiring CADD Researcher | Bridging Biology and Computation

Contact

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Profile

300-level Biochemistry student at UNILAG, actively transitioning from data analytics into bioinformatics and Computer-Aided Drug Discovery (CADD). Bridging biological sciences with data-driven computational methods — homology modeling, molecular docking, and chemoinformatics — supported by a certified data analytics background (SQL, Python, Power BI). Currently building a portfolio of dry-lab computational workflows and preparing for placement in a computational biology research setting.

Computational Biology & CADD Skills

- Homology modeling — PyMOL, SWISS-MODEL
- Molecular docking — AutoDock Vina
- Chemoinformatics — RDKit, Biopython
- Custom WSL Ubuntu environment for computational workflows
- Network pharmacogenomics
- Underlying data skills: Python, R, statistical analysis — carried over from data analytics background

Research & Manuscript Work

- Submitted a manuscript on homology modeling and molecular docking; currently addressing reviewer feedback on visualizations, compound library screening, binding energy interpretation, and scientific writing — active work in progress, not yet published.

Computational Projects

A note on status: projects marked "verify repo link" should only be shared publicly once a corresponding GitHub repository is live and documented to the same standard as the data analytics portfolio (README, methods, results). Claims without a checkable repo are a credibility risk with technical reviewers.

Project	Tools	What it does	Status
AI-Driven Homology & Targeted Drug Discovery	SWISS-MODEL, AutoDock Vina, PyMOL, Python	End-to-end dry-lab pipeline: 3D homology modeling of a protein target followed by molecular docking simulation against candidate ligands.	**Verify repo link before sharing**
Automated Drug Discovery Pipeline	R, RDKit, Virtual Screening	Bridges bio- and chemoinformatics to automate high-throughput virtual screening of small-molecule libraries.	**Verify repo link before sharing**
Agrospectra: NDVI Crop Health Analysis	Python, NumPy, OpenCV	Remote sensing pipeline evaluating agricultural vegetation health — processes	**Live on GitHub**

Project	Tools	What it does	Status
		multi-spectral image bands to calculate NDVI.	

Relevant Experience

SIWES Industrial Trainee

NAFDAC — 2025

- Completed a 3-month industrial attachment; gained exposure to regulatory laboratory environments, quality assurance protocols, and operational workflows relevant to pharmaceutical and drug-discovery contexts.

Assistant Tutor & Community Manager

ULLSSA Skill Up Programme, UNILAG — May 2026 – Present

- Technical mentorship and curriculum delivery for a data analysis track — demonstrates the analytical foundation supporting the CADD transition.

Target Direction

- Actively researching postgraduate and placement options in computational biology, including SIWES and research-track opportunities at institutions with computational biology divisions in Nigeria.

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